Ganghui ZHANG

Oxford, UK   zhangganghui2025@gmail.com

Education

Ph.D.	**Yau Mathematical Sciences Center, Tsinghua University** **Major:** Computational Mathematics **Supervisor:** Prof. Chunmei Su	Sept. 2019 – Dec. 2024
Visiting student	**Department of Mathematics, University of Regensburg** **Host** : Prof. Harald Garcke	Jun. 2024 – Nov. 2024
B.Sc.	**Shandong University** **Major:** Pure and Applied Mathematics	Sept. 2015 – Jun. 2019

Employment

Postdoc	**Mathematical Institute, University of Oxford** **Position:** Postdoctoral Research Assistant **Mentor:** Prof. Kaibo Hu	Sept. 2025 – Present
Postdoc	**Department of Mathematics, University of Edinburgh** **Position:** Postdoctoral Research Assistant **Mentor:** Prof. Kaibo Hu	Apr. 2025 – Aug. 2025

Research interest

- Numerical Analysis and Scientific Computing
- Geometric Partial Differential Equations

Publications

1. H. Garcke, D. Trautwein and **G. Zhang**, Structure-preserving parametric finite element methods for two-phase Stokes flow based on Lagrange multiplier approaches, J. Comp. Phys., 560 (2026), article 114922.
2. H. Garcke, R. Nürnberg, S. Praetorius and **G. Zhang**, Isoparametric finite element methods for mean curvature flow and surface diffusion, J. Comp. Phys., 539 (2025), article 114248.
3. H. Garcke, W. Jiang, C. Su and **G. Zhang**, Structure-preserving parametric finite element method for surface diffusion based on Lagrange multiplier approaches, SIAM J. Sci. Comput., 47 (2025), pp. A1983–A2011.
4. W. Jiang, C. Su and **G. Zhang**, Convergence analysis of three semidiscrete numerical schemes for nonlocal geometric flows including perimeter terms, IMA J. Numer. Anal., 46 (2025), pp. 899–937.
5. W. Jiang, C. Su, **G. Zhang** and L. Zhang, Predictor-corrector, BGN-based parametric finite element methods for surface diffusion, J. Comp. Phys., 530 (2025), article 113901.
6. W. Jiang, C. Su and **G. Zhang**, A second-order in time, BGN-based parametric finite element method for geometric flows of curves, J. Comp. Phys., 514 (2024), article 113220.
7. W. Jiang, C. Su and **G. Zhang**, Stable BDF time discretization of BGN-based parametric finite element methods for geometric flows, SIAM J. Sci. Comput., 46 (2024), pp. A2874–A2898.
8. W. Jiang, C. Su and **G. Zhang**, A convexity-preserving and perimeter-decreasing parametric finite element method for the area-preserving curve shortening flow, SIAM J. Numer. Anal., 61 (2023), pp. 1989–2010.

Preprints

1. B. D. Andrews, P. E. Farrell and **G. Zhang**, Arbitrary-order structure-preserving discretizations for geometric curvature flows, arXiv:2605.20371.
2. P. E. Farrell, M. He, K. Hu and **G. Zhang**, Global and local helicity-preservation in the finite element discretisation of magnetic relaxation, arXiv:2603.12134.
3. G. Gao, K. Hu, B. Li and **G. Zhang**, Finite element methods for isometric embedding of Riemannian manifolds, arXiv:2602.18722.

Minisymposium talks

1. Finite element methods for isometric embedding of Riemannian manifolds, ESI Workshop, Vienna, Austria, May 4, 2026.
2. Finite element methods for isometric embedding of Riemannian manifolds, TSIMF Conference of Finite Element Tensor Calculus, Sanya, China, January 15, 2026.
3. Structure-preserving parametric finite element method based on Lagrange multiplier approaches, Oxford Numerical Analysis Internal Seminar, Oxford, UK, November 20, 2025.
4. Isoparametric finite element methods for mean curvature flow and surface diffusion, Workshop on Formulations and Discretization of Interface Problems, Institut Mittag-Leffler, Sweden, September 5, 2025.
5. Isoparametric finite element methods for mean curvature flow and surface diffusion, 30th Biennial Numerical Analysis Conference, Glasgow, UK, June 25, 2025.
6. Structure-preserving parametric finite element method for surface diffusion based on Lagrange multiplier approaches, 35th Mathematics Rising Star Scholar Forum, Shandong University, Jinan, China, March 13, 2025.
7. Structure-preserving parametric finite element method for surface diffusion based on Lagrange multiplier approaches, RTG IntComSin Annual Meeting, Ellwangen, Germany, September 30, 2024.
8. A convexity-preserving and perimeter-decreasing parametric finite element method for the area-preserving curve shortening flow, Annual Conference of CSIAM, Kunming, China, October 15, 2023.

Teaching

Finite Element Method (Tutor), Mathematical Institute, University of Oxford — Spring 2026

Complex Analysis (Teaching Assistant), Department of Mathematics, Tsinghua University — Spring 2021

Differential Geometry (Teaching Assistant), Department of Mathematics, Tsinghua University — Fall 2020

Honor

Yau Mathematical Sciences Center Scholarship — Tsinghua University 2023

National Scholarship — Shandong University 2018

Softwares

Programming languages: MATLAB, Python, C++.

Finite element libraries: DUNE, NGSolve, Firedrake.